Yihang Jiao

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Research Interests

Machine learning for biomedical data — segmentation, literature retrieval — and the reliable evaluation of deep models under class imbalance and limited supervision.

Education

University of Illinois Urbana-Champaign Expected Dec 2027
Master of Computer Science — GPA 3.91/4.0 **University of Illinois Urbana-Champaign** May 2025
B.S. in Mathematics and Computer Science — GPA 3.83/4.0

Research Experience

FOCAL Lab, UIUC — Prof. Gagandeep Singh Aug 2024 – Dec 2024
Research Assistant — Reliability of Automated Red-Teaming

- Probed text classifiers by having LLaMA-2 rewrite whole sentences rather than swapping words, then asked what such a pipeline measures: across 32 configurations, rewrites of already-successful attacks fool the classifier only 44.5% of the time, buying fluency (median perplexity 22.9 vs. 522) at the price of half the attacks.
- **Label validity.** A prompt ablation on identical data (119/180 vs. 58/180, $p = 1.7 \times 10^{-10}$) showed the winning prompt wins by instructing the model to change the label: an independently trained referee reproduces 86–87% of those "successes", and requiring the flip to be target-specific **reverses the ranking**, 12/180 against 25/180. Adding the cosine constraint this literature relies on does not change the ranking; only checks independent of the embedding space do, and an NLI entailment test disagrees with cosine on 78% of what cosine accepts.
- **Input validity.** On sentence-pair tasks most probes collapse the two segments and are not well-formed inputs; since malformed probes are scored as failures, leaving them in *understates* attack success (12.5% vs. 41.7%) — the two errors run opposite ways and do not cancel.
- **Generator and measurement validity.** The generator refuses 6.2% of requests at a rate tracking the prompt under study, and its refusals are scored as attacks; my own earlier version never ran the classifier on the generated text. Closes with five numbers a report should carry.

Code: `https://github.com/yhj3/counterfactual-textattack`

Method-Specific Paper Recommendation for Academic Writing Aug 2024 – present
UIUC, CS 397 — with Prof. Hongying Zan; manuscript in preparation

- Extended SPECTER embeddings beyond titles and abstracts to subsection titles and method descriptions, aligned them with sentence-level citation contexts, and built citation graphs over direct and multi-level links; method-specific accuracy rose from 47% to 65% over the baseline, judged by student raters using the system in their own research.
- With Yexin Wu, added an LLM reranking stage that compares candidates pairwise, treating the noisy judgments as a dueling-bandit problem (UCB Elimination, roughly 190 comparisons cut to 44); over 10,395 papers and 497,210 citation contexts, in-link context raised MRR from 0.485 to 0.572 at $K = 5$.

Write-up: `https://yhj3.github.io/projects/scilit-rag.html`

Projects

Brain Tumor Segmentation on Multi-Modal MRI: U-Net vs. TransUNet Aug 2025 – Dec 2025
Independent project

- Built a unified 2D segmentation pipeline over four co-registered MRI modalities (FLAIR, T1, T1-CE, T2): a U-Net, a configurable U-Net, and a ResNet50 + ViT TransUNet train through one entrypoint under the same data path, split rule, and metric, and every checkpoint stores its full configuration and crop box so the model can be rebuilt exactly. Data path: brain-region cropping, empty-slice filtering, per-slice z-score normalization, foreground oversampling; objective: class-weighted cross-entropy plus present-class foreground Dice with warmup.
- Found that deepening the U-Net and switching normalization with the loss untouched dropped foreground Dice from 0.7326 to 0.6323; reworking only the objective and sampler on the same architecture recovered about +7 points (0.7033), so loss and sampling design mattered as much as backbone choice.
- TransUNet reached 0.7575 foreground Dice against 0.7326 for the U-Net under a fixed 15-epoch from-scratch budget (best-epoch validation), with the gain localized to the diffuse subregion (0.7652 vs. 0.6800) and none on compact structures; a variant adding skip-connection self-attention, stronger augmentation, and 224^2 input lowered cross-entropy (0.0455 to 0.0411) but also Dice (0.7347).

Code: `https://github.com/yhj3/brats-unet-vs-transunet`
Weights: `https://huggingface.co/yihangj3/brats-transunet`

VibeRepair+: Retrieval-Augmented Program Repair for Java and Python Jan 2026 – May 2026
CS Software Engineering for GenAI — co-author

- Designed a pluggable retriever framework over a 97,402-example bug-fix corpus in which adding a retrieval strategy takes one Python file and no changes to the serving endpoint, evaluation harness, or metrics; reproducibility enforced by a corpus SHA-256 fingerprint and a locked holdout split.
- Diagnosed the incumbent leave-one-out evaluation as a self-match artifact — every retriever scored a perfect recall@1 while agreeing on only 34.1% of top-1 results — and replaced it with an external-query protocol on 521 real Defects4J bugs, with an operator-aware token-overlap proxy ground truth and a two-stage embedding plus edit-distance leak check.
- Under the corrected protocol, lexical BM25 beat the dense-embedding baseline by 27.8% relative recall@10, inverting the usual dense-beats-lexical assumption; a Reciprocal Rank Fusion hybrid led five of six metrics and recovered 15 top-10 hits missed by both sub-retrievers.

Code: `https://github.com/yhj3/Vibe_Repair_Plus`

Skills

ML: PyTorch, CUDA, HuggingFace Transformers, mixed-precision/multi-GPU training, FAISS, scikit-learn, NumPy, pandas. **Biomedical:** multi-modal MRI, 2D/3D segmentation, Dice under class imbalance, corpus construction. **Programming:** Python, C++, C#, Java, Git, LaTeX.